Interior angles of polygons

1

a 4 triangles

b 720°

2

a 4 triangles

b 720°

3

a

i 180 **ii** 180 **iii** 180

b 540

c $n-2$

d

i No **ii** No **iii** Yes **iv** No

4

	Number of Sides	Least Number of Triangles	Interior angle sum
Pentagon	5	3	540°
Hexagon	6	4	720°
Heptagon	7	5	900°
Octagon	8	6	1080°
Nonagon	9	7	1260°
Decagon	10	8	1440°

5

a 360° **b** 1800° **c** 2160° **d** 2520°

e 3240° **f** 6840°

6

a

i $y = 63^\circ$ **ii** $x = 64^\circ$

b

i $y = 101^\circ$ **ii** $x = 139^\circ$

7 $y = 158^\circ$

8

a 1080° **b** 135°

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9

a $x = 120^\circ$

b $x = 108^\circ$

10

a

i $n = 6$ **ii** Hexagon

b

i $n = 8$ **ii** Octagon

c

i $n = 3$ **ii** Triangle

11 Because then $n = 7.2$, and it is not possible to have 7.2 sides of a polygon.

12 1260°

13 135°

Exterior angles of polygons

14

a

i $x = 73^\circ$ **ii** $a = 88^\circ$ **iii** $b = 92^\circ$ **iv** $c = 107^\circ$

v $d = 73^\circ$ **vi** 360°

b The sum of exterior angles in a quadrilateral is 360°.

15 $y = 98^\circ$

16

a $x = 81^\circ$

b $x = 7^\circ$

17

a 72° **b** 45° **c** 120° **d** 40°

18

a 120° **b** 90° **c** 144° **d** 150°

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